

#### 4.) Explain the concept of Linear Regression and Logistic Regression

##### Linear Regression:

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value. It works by finding the relationship between an independent variable(s) and a dependent variable by fitting a straight line through the data points. The model is used to estimate how changes in input variables affect the output.

##### Logistic Regression:

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of an outcome belonging to a specific category, usually represented as binary values such as Yes/No or 0/1. It uses a logistic (sigmoid) function to classify data into groups.

#### 5.) Mention one use case each of Linear Regression and Logistic Regression

- Linear Regression Use Case:

Predicting car prices based on factors such as brand, engine size, mileage, and year of manufacture.

It predicts numbers like price, weight, temperature, etc.

- Logistic Regression Use Case:

Predicting whether a customer will purchase a car or not based on their income, gender, and previous buying behavior. It predicts categories like yes/no, spam /not spam, etc.